What a Single Correlation Hides: Auditing a Language-Model Judge of Formal Specification Coverage

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Abstract

Lean 4 formalization benchmarks use language-model judges to score whether an agent's theorem statements cover a human-written reference. Such a judge is often justified by one correlation between its scores and human ratings. We audit one such justification, a reported Pearson correlation of 0.61, and show what the single coefficient leaves out. A GPT-5.3-Codex judge scored 75 Lean artifacts from 25 tasks that five human raters had rated under a shared 0–5 coverage rubric. We reanalyze the saved scores with task-level resampling, rank and absolute agreement, and a byte-level check of the judge's inputs. The correlation's 95% task-bootstrap interval is [0.21, 0.82], and Spearman correlation is 0.38. The judge also rates artifacts 0.83 points below the human mean on the 0–5 scale. Against lenient raters the correlation is 0.70; against expert raters it is 0.37. For 24 artifacts the judge's reference theorems differ from those the raters saw. On the other 51, Spearman correlation is 0.56. A supervised score remapping removes the offset but also nearly all association. We recommend that coverage-judge audits report task-level intervals, rank and absolute agreement, the rater target, and whether the judge saw the raters' inputs.

1 Introduction

A Lean file can compile, and even prove every theorem it states, while omitting properties that its source program must satisfy. Benchmarks that ask coding agents to formalize programs therefore score *formal specification coverage*. It measures how fully the agent's theorem statements express the obligations in a human-written reference. VeriBench calls this quantity theorem equivalence (TE1) in its report and theorem coverage (TC1) on its launch page (Miranda et al., 2026). Establishing coverage formally requires proving implications between statements, which is itself a hard proof task. Such benchmarks delegate the coverage score to a language-model judge instead. Because that score is one factor of the benchmark's multiplicative overall score (Miranda et al., 2026), the judge's agreement with expert judgment shapes what the benchmark measures.

Language-model judges are usually justified by their agreement with human ratings (Zheng et al., 2023; Liu et al., 2023; Chiang & Lee, 2023). That agreement is often summarized by a single correlation coefficient. A correlation measures association rather than agreement: a method can correlate with a reference while reading consistently higher or lower (Bland & Altman, 1986). Its value also depends on choices that the coefficient does not record. These include how ratings are grouped, which raters define the target, and whether the judge read the same inputs as the raters.

We treat a reported judge–human correlation as a measurement to audit rather than a number to accept. Our object is the evidence behind one benchmark launch claim. VeriBench reports that a GPT-5.3-Codex judge given the raters' rubric reaches Pearson correlation $r = 0.6137$ against its human ratings, with weaker rank correlations (Miranda et al., 2026). It presents this result as an alignment check, not as evidence that its deployed coverage scorer was fitted to human ratings. We ask what the reported number establishes about the judge.

We reanalyze the archived per-artifact scores of that judge and the human ratings it was compared with, without new model calls or new ratings. The data are 75 Lean artifacts from 25 tasks. Five raters rated

them on a 0–5 coverage rubric, and the judge scored each artifact three times. We complement the reported coefficient with intervals that resample tasks and with rank and absolute agreement. We also separate within-task from between-task correlation, vary the rater target, and hash-compare the judge's inputs with the rated files (Figure 1).

The reported correlation reproduces exactly, but its 95% interval over tasks is [0.213, 0.821], and Spearman correlation is 0.375 [0.013, 0.668]. The judge rates artifacts 0.825 points below the human mean on the 0–5 scale. Its absolute-agreement intraclass correlation, 0.429, therefore falls well below its consistency intraclass correlation, 0.613. After centering scores within each task, Pearson correlation drops to 0.358 [0.055, 0.635]. The Spearman correlation computed inside each task averages only 0.255 [−0.073, 0.579]. The same judge scores have Pearson correlation 0.702 with the mean of lenient raters and 0.372 with the mean of expert raters. For eight of the 25 tasks, the judge's reference theorems differ from those the raters saw. On the other 17 tasks, Spearman correlation is 0.565 [0.197, 0.800], but the downward offset remains. A supervised isotonic remapping removes the offset but leaves Spearman correlation at 0.046, and its absolute error is close to that of a constant prediction.

These results neither validate nor reject the judge; they show that the reported coefficient alone cannot tell which. They also concern a different judge from the one that produced the benchmark's reported coverage scores, which used another model and rubric (Miranda et al., 2026). We make the following contributions:

1. We reconstruct the evidence behind a reported judge–human correlation for formal specification coverage and reproduce it exactly from archived receipts.
2. We show that task-level uncertainty, rank and absolute agreement, the choice of rater target, and mismatched judge inputs each change what the correlation supports, with task-bootstrap intervals for every statistic we derive against the pooled target.
3. We derive a short reporting checklist for audits of coverage judges from these findings.

2 Audit Setup

The audit reuses archived ratings and judge scores; it adds no model calls and no ratings.

2.1 Tasks and artifacts

The rating pool contains 25 VeriBench tasks, five from each of the benchmark's security, real-code, easy, computer-science and HumanEval-derived task families. Each task pairs a Python program with a human-written reference Lean 4 file (de Moura & Ullrich, 2021) that states the program's required properties as theorems. Each task has three *artifacts*, candidate Lean files sampled independently from `claude-3-7-sonnet-20250219` at temperature 0.7, one zero-shot call each, without tools or compiler feedback. Generation receipts confirm this for 74 of the 75 artifacts; one receipt is missing. The three artifacts of a task are samples from one model, not the outputs of three different agents. The generation prompt allowed unfinished proofs, and 73 of the 75 artifacts contain at least one `sorry` placeholder. Coverage concerns theorem statements, so unfinished proofs do not lower it by definition. The raters and the judge nevertheless saw the full files, proofs included.

2.2 Human ratings and targets

Five raters, expert and collaborator volunteers, rated coverage on a 0–5 scale after reading the Python program, the reference file and the artifact (Miranda et al., 2026). They were instructed to work independently, to judge statement coverage rather than proof style, and not to use language models or automated judges; compliance was not recorded. The rubric assigns a base score from 5, all reference theorems covered, to 0, vacuous or broken, and lowers it by up to two points when irrelevant theorems dominate (Appendix A). Ratings come from seven files: participant 1 rated the artifacts twice under an earlier version of the prompt

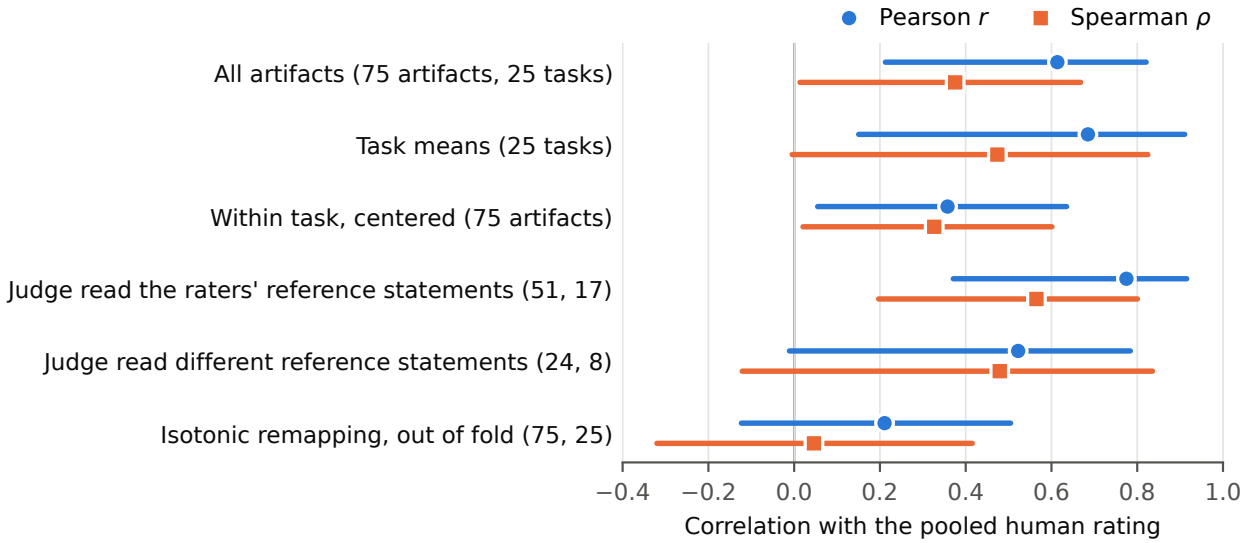


Figure 1: **The same judge scores give correlations from near zero to 0.77, depending on how agreement is computed.** Pearson r (circles) and Spearman ρ (squares) between the GPT-5.3-Codex judge and the pooled human rating, with 95% percentile intervals from 10,000 bootstrap draws over tasks. Parentheses give artifacts and tasks. The fourth and fifth rows split the artifacts by whether the judge read the same reference theorem statements as the raters (Section 3.5). The last row applies an isotonic mapping fitted on the other fold of a two-fold task split. Only the point estimates of the first and last rows appear in the original receipt; every interval and every other row is a post hoc analysis of the archived scores.

and once under the final prompt, and participants 2–5 rated once each. Under the final prompt the five raters supplied 345 of 375 possible ratings; participant 3 rated 45 of the 75 artifacts.

We compare the judge with four targets, each an artifact-level mean of 0–5 ratings. The *pooled target* averages all available ratings for the artifact across the seven files; the reported correlation uses it. It counts participant 1 up to three times. The *lenient group* averages participant 1's two earlier-prompt passes and participants 4 and 5, and the *expert group* averages participant 1's final-prompt ratings and participants 2 and 3. The group names describe the groups' score levels, not the quality of their raters (Miranda et al., 2026). The *five-rater target* averages one final-prompt rating per rater. Every artifact has at least one rating under every target.

2.3 The audited judge

The judge is `gpt-5.3-codex`, run through the Codex command-line interface (`codex exec`) with reasoning effort `xhigh`. Its system prompt restates the raters' rubric, with the same base-score definitions, checklist and spam adjustment, and asks for a structured answer (Appendix A). Its user prompt contains the full Python program, the full reference file and the full artifact. The judge scored each artifact three times, and an artifact's judge score is the mean of the three final scores. The run, made in May 2026, scored all 75 artifacts by model calls, with no failed calls.

Two earlier configurations, which saw only extracted theorem statements, reached Pearson correlations of 0.333 and 0.406 with the same pooled target. The full-file configuration was retained because it agreed best with that target. Its agreement is therefore a result selected on the same ratings it is evaluated against, and no separate test set exists.

This judge did not produce VeriBench's reported coverage scores. Those came from `claude-sonnet-4-6`, which rated each artifact's theorem statements from 0 to 10 under an equivalence-style rubric and took the

Statistic	Raw judge	Isotonic remapping, out of fold
Pearson r	0.614 [0.213, 0.821]	0.211 [−0.123, 0.505]
Spearman ρ	0.375 [0.013, 0.668]	0.046 [−0.320, 0.415]
Kendall τ_b	0.300 [0.021, 0.545]	0.042 [−0.232, 0.320]
$r - \rho$	0.238 [0.061, 0.408]	0.165 [−0.027, 0.342]
ICC(A,1), absolute agreement	0.429 [0.110, 0.662]	0.167 [−0.102, 0.379]
ICC(C,1), consistency	0.613 [0.206, 0.819]	0.165 [−0.108, 0.387]
Judge mean	2.382 [2.067, 2.680]	3.190 [3.060, 3.310]
Pooled human mean	3.207 [2.880, 3.499]	3.207 [2.880, 3.499]
Mean bias, judge minus human	−0.825 [−1.082, −0.594]	−0.018 [−0.332, 0.301]
Mean absolute error	0.889 [0.695, 1.122]	0.628 [0.432, 0.848]

Table 1: **The judge correlates moderately with the pooled human target but rates artifacts 0.825 points lower; remapping removes the offset and the association.** Agreement with the pooled target on 75 artifacts from 25 tasks; means, bias and error are on the 0–5 scale. Brackets give 95% percentile intervals from 10,000 bootstrap draws over tasks. The receipt's p-values for r, ρ and τ_b, which treat artifacts as independent, are 4.78×10^{-9}, 9.03×10^{-4} and 4.49×10^{-4} for the raw judge and 0.069, 0.694 and 0.637 after remapping. Predicting the pooled human mean for every artifact gives a mean absolute error of 0.643 in sample and 0.705 when each fold uses the other fold's mean.

median of three calls (Miranda et al., 2026). That evaluation did not report the deployed judge's agreement with these human ratings, so the present audit does not measure it.

2.4 Statistics

We report Pearson correlation r, Spearman correlation ρ (Spearman, 1904) and Kendall correlation τ_b (Kendall, 1938) between judge scores and a target. The receipt's p-values test zero association and treat the 75 artifacts as independent. For agreement in level, we report the mean bias, judge minus target, so that a negative value marks a stricter judge, and the mean absolute error, both on the 0–5 scale. We also report two intraclass correlations, ICC(A,1) for absolute agreement and ICC(C,1) for consistency (Shrout & Fleiss, 1979; McGraw & Wong, 1996). The gap between them reflects a constant offset that correlations ignore (Appendix B).

The three artifacts of a task share a program and a reference, so tasks, not artifacts, are the independent units. Every interval is a 95% percentile interval from 10,000 bootstrap draws that resample the 25 tasks with replacement, each keeping its three artifacts (Efron & Tibshirani, 1993; Field & Welsh, 2007). Differences between two statistics, such as $r - \rho$, are computed within each draw. We also correlate task means (25 points) and scores centered within each task, which separate between-task from within-task agreement. The Spearman correlation of centered scores still ranks residuals across all 75 artifacts, so for ordering within a task we also average the Spearman correlation computed inside each task, over the tasks where neither score is constant. For absolute error, we compare each scorer with a constant prediction of the human mean and bootstrap the paired difference in mean absolute error.

For the input check, we compare SHA-256 hashes of the reference and artifact files in the judge run's commit with the hashes that a later audit of the rating pool recorded for the files as rated. That audit also recorded whether each reference's theorem statements changed after rating. For the remapping analysis, we reproduce the receipt's procedure: isotonic regression (Barlow et al., 1972) of the target on the judge score, fitted on the tasks of one fold of a two-fold task split and applied to the other fold. The bootstrap holds the two fitted maps fixed. Beyond the receipt's three correlations for each scorer, their p-values and its rater-group table, every statistic in this paper is a post hoc analysis of the archived scores, computed for this audit.

Target	Ratings averaged	Artifacts	Human mean	r	ρ	τ_b	Judge − human
Pooled	all seven files	75	3.207	0.614	0.375	0.300	−0.825
Lenient group	P1 earlier (×2), P4, P5	75	3.573	0.702	0.603	0.486	−1.191
Five raters	P1–P5, final prompt	75	2.989	0.500	0.339	0.267	−0.607
Expert group	P1 final, P2, P3	75	2.631	0.372	0.300	0.218	−0.249
Participant 1	final prompt	71	2.690	0.275	0.209	0.151	−0.329
Participant 2		75	2.533	0.313	0.269	0.208	−0.151
Participant 3		44	2.227	0.742	0.442	0.359	+0.303
Participant 4		75	3.867	0.520	0.404	0.336	−1.484
Participant 5		75	3.133	0.474	0.300	0.247	−0.751

Table 2: **Correlation and bias move in opposite directions across rater targets.** Agreement of the same judge scores with different human targets; means and differences are on the 0–5 scale. P1–P5 denote participants; "earlier" marks participant 1's two passes under an earlier prompt. Correlations and human means below the pooled row are point estimates from the run's receipt, which reports no intervals; we reproduced each from the rating files. The last column subtracts the human mean from the judge's mean over the same artifacts. Participant 1 has 71 artifacts because three rows name an older reference file and one row is malformed; participant 3's file held 44 ratings at the time of the run.

3 Results

3.1 The reported correlation is reproducible but imprecise

The audit reproduces the reported coefficients to within 10^{-12}: $r = 0.614$, $\rho = 0.375$ and $\tau_b = 0.300$ over the 75 artifacts (Table 1). A test that treats the artifacts as independent gives $p = 4.78 \times 10^{-9}$ for the Pearson correlation. That p-value suggests far more precision than the data hold, because only the 25 tasks are independent. Resampling tasks gives a 95% interval of [0.213, 0.821] for r. The data are thus consistent with anything from a weak to a strong linear association.

3.2 Rank and level agreement are weaker than linear association

Spearman correlation is 0.375 [0.013, 0.668], and Kendall correlation is 0.300 [0.021, 0.545]. The paired difference $r - \rho = 0.238$ [0.061, 0.408] excludes zero, so the judge orders artifacts less consistently than its Pearson correlation suggests.

The judge's mean score is 2.382, against a pooled human mean of 3.207, a mean bias of −0.825 [−1.082, −0.594] on the 0–5 scale. The judge is thus stricter than the raters behind this target. Consistency, which ignores a constant offset, reaches ICC(C,1) = 0.613 [0.206, 0.819], whereas absolute agreement reaches only ICC(A,1) = 0.429 [0.110, 0.662]. The judge's mean absolute error, 0.889 [0.695, 1.122], is larger than that of predicting the pooled human mean for every artifact: 0.643 in sample and 0.705 when each fold of the task split uses the other fold's mean. Against that out-of-fold constant, the paired difference is 0.184 [−0.098, 0.440], so the raw judge is not more accurate in level than a constant prediction. Read as ratings on the raters' scale, the judge's scores are biased low, and the positive correlation does not reveal it.

3.3 Most of the association lies between tasks

Averaging the three artifacts of each task gives $r = 0.685$ [0.150, 0.911] over the 25 task means. Centering scores within each task removes differences between tasks and leaves $r = 0.358$ [0.055, 0.635] (Figure 1). The Spearman correlation of the centered scores is 0.327 [0.021, 0.601], but it still compares residuals across tasks. Computed inside each task, over the 20 tasks where neither score is constant, Spearman correlation averages 0.255 [−0.073, 0.579]. The judge therefore tracks which tasks received higher ratings more closely than it orders the three artifacts of one task. Comparisons of different outputs for the same task depend on this weaker within-task ordering.

Artifacts	Count (tasks)	Pearson r	Spearman ρ	ICC(A,1)	Mean bias
All	75 (25)	0.614 [0.213, 0.821]	0.375 [0.013, 0.668]	0.429 [0.110, 0.662]	−0.825 [−1.082, −0.594]
Reference statements match	51 (17)	0.775 [0.371, 0.915]	0.565 [0.197, 0.800]	0.617 [0.216, 0.794]	−0.587 [−0.760, −0.393]
and unchanged while rated	36 (12)	0.764 [0.357, 0.914]	0.632 [0.237, 0.863]	0.624 [0.216, 0.807]	−0.623 [−0.851, −0.345]
Reference statements differ	24 (8)	0.522 [−0.011, 0.784]	0.479 [−0.121, 0.836]	0.265 [−0.002, 0.572]	−1.330 [−1.838, −0.866]

Table 3: **Where the judge read the raters' reference theorems, rank and absolute agreement rise, but the judge remains stricter than the raters.** Agreement with the pooled target, split by whether the reference file in the judge run states the same theorems as the reference the raters rated. "Unchanged while rated" further drops five tasks whose references changed during the rating window. The mismatched row includes one task whose judge-run reference matches neither recorded version and states a different number of theorems. Brackets give 95% percentile intervals from 10,000 bootstrap draws over tasks; bias is on the 0–5 scale. These subsets were defined after the scores were seen.

3.4 The rater target changes both correlation and bias

The same judge scores have Pearson correlation 0.702 with the lenient group's mean and 0.372 with the expert group's mean (Table 2). Bias moves the other way: the judge rates 1.191 points below the lenient group but only 0.249 below the expert group. Among the four group targets, the one that maximizes the judge's correlation is thus the one whose score level it matches worst. Against single raters, Pearson correlation ranges from 0.275 to 0.742, on 44 to 75 artifacts. The judge's mean ranges from 1.484 points below participant 4 to 0.303 points above participant 3, so whether the judge counts as strict depends on the rater. For comparison, each expert's ratings correlate with the mean of the other available experts' ratings at an average Spearman correlation of 0.764. The judge reaches $\rho = 0.300$ against the mean of all three experts, a target that averages more ratings and is therefore less noisy than the other-expert means in that comparison. Neither figure carries an interval, but a gap this large indicates that the judge's rank agreement with the expert group falls well short of the experts' agreement with one another.

3.5 The judge did not always read the raters' references

All 75 artifacts in the judge run are byte-identical to the artifacts that were rated. The references are not. For 14 tasks the judge read the reference recorded as rated, and for 3 tasks a later file with the same theorem statements. For 7 tasks it read a later reference whose theorem statements had changed, and for 1 task a third version that states 7 theorems where the rated reference stated 5. In these 8 tasks, 24 artifacts, the judge's reference theorems therefore differ from the rated ones. They include all five tasks whose references, when rated, held only placeholder theorems; the raters scored those artifacts against the placeholders, while the judge scored them against later references with different statements. On these 24 artifacts the raters' mean is 3.553 and the judge's is 2.222, a mean bias of −1.330 [−1.838, −0.866] (Table 3).

On the 51 artifacts whose reference statements match, $r = 0.775$ [0.371, 0.915], $\rho = 0.565$ [0.197, 0.800] and ICC(A,1) = 0.617 [0.216, 0.794]. Five of the matched tasks had references that changed during the rating window, so the version each rater opened is uncertain for them. Dropping those tasks leaves 36 artifacts with $\rho = 0.632$ [0.237, 0.863]. The judge stays stricter on matched inputs, with a mean bias of −0.587 [−0.760, −0.393]. Restricting to matched inputs therefore removes much of the weak rank agreement but not the judge's strictness. Because the five placeholder tasks form the whole security family, the matched subset contains no security tasks, so reference mismatch and task family are confounded here.

3.6 Remapping removes the offset and the association

An earlier analysis, whose scorer package is no longer available, reported $r = 0.703$ for a GPT-5.3-Codex scorer after an isotonic mapping fitted across a two-fold task split, against one rater's averaged earlier-

prompt passes on 72 artifacts. The audited run was an attempt to reproduce that result. Applying the same procedure to the audited judge gives out-of-fold $r = 0.211$ [−0.123, 0.505], $\rho = 0.046$ [−0.320, 0.415] and $\tau_b = 0.042$ [−0.232, 0.320] (Table 1). The remapping preserves order within each fold, where Spearman correlation is 0.596 and 0.238, but each fold is scored by the map fitted on the other fold. The fold whose ratings average 2.966 therefore receives predictions averaging 3.376, while the fold whose ratings average 3.468 receives 2.989, and the first fold gets only two distinct values. Pooling the two folds scrambles the order between them. The remapped scores are nearly unbiased overall, at −0.018 [−0.332, 0.301], and their mean absolute error, 0.628, is lower than the raw judge's 0.889. That error is close to a constant prediction's: it lies 0.077 [−0.178, 0.020] below the out-of-fold constant, an interval that includes zero. The remapping therefore lowers error by pulling scores toward the human mean rather than by ordering artifacts better. Lower absolute error therefore does not imply a better judge; error and rank agreement have to be read together.

4 What the Audit Does Not Establish

The deployed scorer. The audited judge differs from the judge behind the benchmark's reported coverage scores in model, rubric, input scope and rating scale (Section 2.3). Our findings bound what the reported correlation says about the audited judge; they do not transfer numerically to the deployed scorer, whose own agreement with these ratings needs a separate audit.

Selection on the evaluation ratings. The full-file configuration was chosen because it agreed best with the same ratings, among three configurations (Section 2.3). Its agreement is a selected value on those ratings, which typically overstates agreement on new ones (Cawley & Talbot, 2010). No untouched test set exists for this judge.

Reconstructed inputs. The input check compares files in the judge run's commit, the best surviving record of what the judge read; uncommitted changes at run time cannot be excluded. For five tasks, references changed during the rating window, so the version each rater opened is itself uncertain. The matched and mismatched subsets were defined after the scores were seen, and they confound reference mismatch with task family.

A narrow population of artifacts. The pool holds 25 tasks and three samples per task from one older model, generated without tools or compiler feedback. Agreement on these artifacts does not establish agreement on outputs of current coding agents, which may differ in length, structure and proof completeness.

Target construction. The pooled target counts participant 1 up to three times, two of them under an earlier prompt. The rater-group rows come from the run's receipt without intervals. That receipt matched rating rows to artifacts by both the reference and the artifact file name, whereas the per-artifact receipt matched by artifact file name alone; we reproduced both exactly from the rating files. The two rules keep different rows, so the group receipt's own pooled row, $r = 0.618$ with human mean 3.175, differs slightly from the per-artifact receipt's $r = 0.614$ with human mean 3.207, which we use. The same matching leaves participant 1 with 71 artifacts, and participant 3's file held 44 ratings at the time of the run, against 45 in a later count of the same study. Historical raters were told not to use language models, but compliance was not recorded.

Repeatability. The per-artifact receipt stores the mean of three judge calls, not the individual calls. We therefore cannot separate disagreement caused by run-to-run variation of the judge from systematic disagreement with the raters.

Agreement is not correctness. Agreement with human raters measures fidelity to a rating procedure, not the logical relation between theorem statements. Raters can share errors, and no implication between a reference and an artifact was proved here.

5 Related Work

Language-model judges and human agreement. Zheng et al. (2023) compare language-model judges with human preferences on conversational responses and document position, verbosity and self-enhancement biases. Liu et al. (2023) report rank correlations between a prompted judge and human judgments of generated text. Chiang & Lee (2023) ask whether language-model evaluation can replace human evaluation by comparing both on the same rating tasks. These studies concern natural-language outputs rated on human preference scales; coverage judging instead compares theorem statements against a formal reference. We follow their practice of measuring agreement with humans, and examine which statistics, targets and inputs that measurement depends on.

Association versus agreement. Bland & Altman (1986) showed that two measurement methods can be highly correlated yet disagree systematically, and proposed analyzing their differences directly. Intraclass correlations separate consistency from absolute agreement for paired ratings (Shrout & Fleiss, 1979; McGraw & Wong, 1996), and the concordance correlation coefficient combines association with agreement in level (Lin, 1989). We report the two intraclass correlations together with mean bias, so that a constant offset between judge and raters stays visible.

Clustered data and selection. When observations share a cluster, resampling whole clusters gives more honest uncertainty than resampling observations (Field & Welsh, 2007). Here the cluster is a task, whose three artifacts share a program and a reference. Choosing among configurations on the evaluation data biases the resulting performance estimate (Cawley & Talbot, 2010), which applies to the selected judge configuration we audit.

Formalization benchmarks. Autoformalization translates informal mathematics or programs into formal statements (Wu et al., 2022). VeriBench evaluates coding agents that translate Python programs into Lean 4 artifacts and scores their theorem statements with a language-model coverage judge (Miranda et al., 2026). The judge examined here is the one whose human agreement that benchmark reports; we do not evaluate the benchmark or its agents.

6 Recommendations and Conclusion

Each check in this audit changed what the reported correlation supports. We therefore recommend that an audit of a coverage judge report the following, each with the reason it matters here.

1. **Intervals over tasks.** Resample tasks rather than artifacts; with 25 tasks, the interval for r spans 0.213 to 0.821 although the independence-based p-value is 4.78×10^{-9}.
2. **Rank and absolute agreement beside Pearson correlation.** Report Spearman and Kendall correlations, ICC(A,1) with ICC(C,1), mean bias and mean absolute error with a constant baseline; here they reveal weaker ordering and a judge that rates 0.825 points low.
3. **Agreement within tasks.** When the judge compares outputs for the same task, report within-task agreement; here the centered Pearson correlation is 0.358 and the average within-task Spearman correlation 0.255, against 0.685 for task means.
4. **The rater target.** Name the raters behind the target and report alternative targets when raters differ in level; here Pearson correlation ranges from 0.372 to 0.702 across rater groups, and the judge's mean ranges from 1.484 points below one rater to 0.303 points above another.
5. **The judge's inputs.** Hash-compare every file the judge reads with the file the raters read before scoring; here 11 of 25 reference files differed in bytes, and 8 stated different theorems from the rated versions.
6. **The deployed configuration.** Audit the judge that produces the benchmark's scores, on ratings not used to choose it.

These items need no new ratings or model calls when the scores are saved per artifact with their task identifiers and input hashes.

The reported correlation of 0.61 is reproducible. On its own, however, it supports only an uncertain, moderate association between one judge and one pooled target. The full audit shows a judge that is stricter than every rater group, orders artifacts within a task weakly, and agrees better when it reads the raters' references. Reporting judge agreement as an audit rather than as one coefficient makes these properties visible to the benchmark's users.

Broader Impact Statement

This work reanalyzes archived ratings and judge scores; it collects no new data from people and deploys no model. We report only aggregate statistics and refer to raters by participant number. Fuller audits of benchmark judges can prevent overstated claims about the systems those benchmarks rank. They also add reporting work, and a judge that passes such an audit on one population of outputs still needs re-auditing on another.

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A Rating Rubric and Judge Prompts

The raters' final prompt, as published with the benchmark (Miranda et al., 2026), reads as follows; we added line breaks only.

```
Rate how well agent Lean theorem statements cover gold Lean theorem statements.
Use Python source + gold Lean file + agent Lean file.
Scope: statement coverage only (not proof style/quality).
Coverage allows the same theorem, logical equivalence, an agent theorem that implies
the gold theorem, or a small lemma set that recovers the gold theorem.

Checklist:
  A. key theorems covered?
  B. pre/postconditions preserved or strengthened?
  C. theorem spam?
  D. if spam, handful or dominates?

Base score 0--5: 5 all covered, 4 mostly covered, 3 partial with important gaps,
2 weak (< half), 1 none covered but relevant non-vacuous theorem, 0 vacuous/broken.
Spam adjustment: C=No no change; C=Yes and D=handful lower up to 1;
C=Yes and D=dominates lower up to 2.
```

The audited judge received the following system prompt, verbatim.

```
You are a Lean 4 theorem-coverage judge.

Rate how well agent Lean theorem statements cover gold Lean theorem statements.
Use the Python source, gold Lean file, and agent Lean file.

Scope: statement coverage only, not proof style or proof quality.
Coverage allows the same theorem, a logically equivalent theorem, a stronger
theorem implying the gold theorem, or a small lemma set that recovers the gold
theorem. Ignore theorem names unless they reveal semantic mismatch.

Base score 0--5:
5 = all gold guarantees covered
4 = mostly covered; only 1-2 minor gaps
3 = partial; at least one important gap
2 = weak; only a few gold theorems covered (< half)
```

```
1 = no gold theorem covered, but at least one non-vacuous relevant theorem
0 = vacuous/broken; no meaningful relevant theorem coverage

Spam adjustment:
- no spam: no change
- handful of spam theorems: lower by up to 1
- spam dominates: lower by up to 2

Return JSON only with this schema:
{
  "A_key_gold_theorems_covered": "Y" | "N",
  "B_pre_post_preserved_or_strengthened": "Y" | "N",
  "C_theorem_spam": "Y" | "N",
  "D_spam_severity": "N/A" | "handful" | "dominates",
  "base_score_0_5": <number>,
  "spam_adjustment": <number>,
  "final_score_0_5": <number>,
  "confidence": "Low" | "Med" | "High",
  "rationale": "<2-4 sentences naming covered and missing gold content>"
}
```

Its user prompt repeats the task statement, embeds the three files in fenced code blocks under the headings `Python Source`, `Gold Lean File` and `Agent Lean File`, and closes with the checklist, the base-score scale and the spam adjustment above. The run's parser reads `final_score_0_5` and falls back to other score fields only when that field is missing or outside 0–5. The artifact's judge score is the mean of the three parsed scores.

B Statistical Definitions and Additional Results

Intraclass correlations. Let y_{i1} be the judge score and y_{i2} the target for artifact $i = 1, \ldots, n$, with $k = 2$ columns. Let MS_R, MS_C and MS_E be the mean squares for artifacts, columns and residual error of the two-way analysis of variance. Following McGraw & Wong (1996),

$$\mathrm{ICC}(C,1) = \frac{\mathrm{MS}_R - \mathrm{MS}_E}{\mathrm{MS}_R + (k-1)\,\mathrm{MS}_E}, \qquad \mathrm{ICC}(A,1) = \frac{\mathrm{MS}_R - \mathrm{MS}_E}{\mathrm{MS}_R + (k-1)\,\mathrm{MS}_E + \frac{k}{n}\,(\mathrm{MS}_C - \mathrm{MS}_E)}. \tag{1}$$

ICC(A,1) is the single-score absolute-agreement coefficient, called ICC(2,1) by Shrout & Fleiss (1979), and ICC(C,1) is their consistency coefficient ICC(3,1). A constant offset between the columns raises MS_C and lowers only ICC(A,1).

Resampling. Each bootstrap draw samples 25 tasks with replacement and keeps all three artifacts of each sampled task. All statistics in a row are computed on the same draws, so differences such as $r - \rho$ are paired. For subsets, the draw samples tasks within the subset. Intervals are the 2.5th and 97.5th percentiles of 10,000 draws with seed 20261112; no draw was degenerate. For the average within-task Spearman correlation, each draw averages over its sampled tasks where neither score is constant. The isotonic predictions are the receipt's out-of-fold values, so their intervals hold the two fitted maps fixed.

Additional subsets. Table 4 adds two subsets defined by reference flags recorded at rating time. Excluding the five placeholder-reference tasks gives results close to the matched-input subset, because those tasks are also the largest group of mismatched inputs.

Input-audit counts. Table 5 lists how the reference in the judge run relates to the reference as rated. All 75 artifact files in the run match the rated artifacts byte for byte.

Artifacts	Count (tasks)	Pearson r	Spearman ρ	Kendall τ_b	ICC(C,1)	Mean absolute error
All	75 (25)	0.614 [0.213, 0.821]	0.375 [0.013, 0.668]	0.300 [0.021, 0.545]	0.613 [0.206, 0.819]	0.889 [0.695, 1.122]
No placeholder reference	60 (20)	0.778 [0.455, 0.901]	0.618 [0.320, 0.811]	0.492 [0.250, 0.677]	0.772 [0.438, 0.898]	0.677 [0.563, 0.795]
and unchanged while rated	45 (15)	0.762 [0.437, 0.896]	0.670 [0.351, 0.852]	0.524 [0.260, 0.712]	0.758 [0.420, 0.890]	0.736 [0.615, 0.863]
Reference statements match	51 (17)	0.775 [0.371, 0.915]	0.565 [0.197, 0.800]	0.451 [0.153, 0.672]	0.765 [0.351, 0.910]	0.674 [0.544, 0.809]
and unchanged while rated	36 (12)	0.764 [0.357, 0.914]	0.632 [0.237, 0.863]	0.493 [0.171, 0.732]	0.755 [0.340, 0.907]	0.746 [0.597, 0.901]
Reference statements differ	24 (8)	0.522 [−0.011, 0.784]	0.479 [−0.121, 0.836]	0.371 [−0.099, 0.709]	0.522 [−0.009, 0.778]	1.344 [0.885, 1.845]

Table 4: **Subsets whose references match the rated versions agree better than the full pool, but every subset keeps a mean absolute error of at least 0.674.** Agreement with the pooled target for subsets defined after the scores were seen; error is on the 0–5 scale. "No placeholder reference" drops the five tasks whose references held only placeholder theorems when rated, which are the five security tasks. Brackets give 95% percentile intervals from 10,000 bootstrap draws over tasks.

Reference in the judge run	Statements later changed?	Changed while rated?	Artifacts
Same bytes as rated	no	no	27
Same bytes as rated	no	yes	12
Same bytes as rated	yes	yes	3
Later file, same statements	no	no	9
Later file, changed statements	yes	no	21
Third version	yes	no	3

Table 5: **The judge read the rated reference statements for 51 of 75 artifacts.** "Statements later changed?" records whether the reference's theorem statements differ between the rated version and a later snapshot, regardless of which version the judge read. The 21 artifacts with a later, changed reference include the 15 artifacts of the five placeholder-reference tasks. The first four rows form the matched subset of Table 3.